

Hieu Nguyen Nguyen

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EDUCATION

International University - VNU HCMC

Bachelor of Data Science

Ho Chi Minh City, Vietnam

2021 – 2026

Le Khiet High School for the Gifted

Informatics Major

Quang Ngai Province, Vietnam

2018 – 2021

EXPERIENCE

HDBank

Data Engineer Intern

May 2025 – Nov 2025

Ho Chi Minh City, Vietnam

- Built automated data pipelines using Python and **n8n** for enriching banking data for financial analysis purposes.
- Implemented data validation systems to ensure high quality data with data integrity, as well as optimized Elasticsearch clusters for **1 million of financial data**.
- Developed monitoring pipelines using Fluent Bit, Prometheus, and Grafana to monitor the health of data pipelines and **achieve 99% pipeline uptime**.
- Created utility tools for testing and deploying the banking data pipelines.
- Worked closely with financial stakeholders** to design data processing pipelines based on business requirements.

International University - VNU HCMC

Team Leader

Jan 2024 – Jan 2025

Ho Chi Minh City, Vietnam

- Worked as a team of 5 in providing software solutions through the use of object-oriented programming (OOP) and **relational databases**, linking application functionality to business requirements.
- Created relational database schemas and implemented **the 2NF** to eliminate data redundancy and ensure consistency of the data relationships and elimination of insertion, update and deletion anomalies.

PROJECTS

High Throughput E-commerce Data Pipeline

Oct 2025 – Dec 2025

Python, Asyncio, ETL

- Developed an asynchronous data ingestion pipeline to collect and preprocess product data from e-commerce APIs, achieving throughput of **200,000 records per hour**.
- Developed a modular architecture having distinct modules for loading, fetching, transforming, and writing that generates structured datasets for further usage in machine learning pipelines. **Validated of the dataset, automated check pointing for capturing failure records**.
- Performed automated collection and preprocessing of the data, thereby **achieving 90% reduction** in the time taken for data collection.

Glamira E-commerce Analytics Pipeline

Jan 2026 – Mar 2026

Python, GCP, MongoDB, BigQuery, dbt, Looker

- Built an end-to-end ETL pipeline to process **41 million e-commerce events (Glamira)**, extracting data from MongoDB and loading it into BigQuery through Google Cloud Storage and Cloud Functions.
- Developed Python workflows to **enrich event data** with scraped product attributes and IP geolocation information for product and geographic analysis.
- Used dbt to build **dimensional models** following a **star schema**, **validate transformed data**, and **anonymize personally identifiable information**.
- Integrated the data warehouse with Looker dashboards to analyze product performance, conversion rates, and geographic trends.

Personalized Decision Support System Using Banking Revenue Data

Sep 2025 – Sep 2026

Python, Scikit-learn, XGBoost, LightGBM, Streamlit, SHAP

- Automated banking data preparation, **engineered predictive features**, and applied recursive feature elimination with cross-validation (RFECV) to select model inputs.
- Built a stacking ensemble combining XGBoost and LightGBM, using Optuna for hyperparameter tuning and SMOTE to address class imbalance, **achieving a ROC AUC above 0.97**.
- Developed an interactive Streamlit application for risk assessment, integrating **SHAP explanations** to show how individual features influenced predictions.

TECHNICAL SKILLS

Languages: Python, Java, SQL, Bash/Shell (Linux/Unix).

Cloud Platforms & DevOps: Google Cloud Platform (GCS, Compute Engine, BigQuery), Git, Docker.

Big Data Engineering: Apache Spark, PySpark, Apache Kafka.

ETL/ELT & Data Modeling: dbt, Airflow, Star Schema, Data Pipeline Design, ETL/ELT Pipelines.

Database: PostgreSQL, MySQL, SQL Server, MongoDB, Elasticsearch.